

Orthogonality

T.T.

Orthogonality decomposes vectors and subspaces. Projections lead to orthogonal bases and least squares; the four fundamental subspaces describe the complete geometry of $Ax = b$.

ORTHOGONAL COMPLEMENTS

$$V \perp W \iff v^T w = 0 \quad \forall v \in V, w \in W, \quad S^\perp = \{x : x^T s = 0 \quad \forall s \in S\}.$$

$$\mathcal{R}(A) = \mathcal{N}(A)^\perp, \quad \mathcal{N}(A) = \mathcal{R}(A)^\perp, \quad \mathcal{C}(A) = \mathcal{N}(A^T)^\perp.$$

$$\mathbb{R}^n = \mathcal{R}(A) \oplus \mathcal{N}(A), \quad x = x_r + x_n.$$

PROJECTIONS

Projection onto a line:

$$\text{proj}_a b = \frac{a^T b}{a^T a} a.$$

$$b = p + e, \quad p = \text{proj}_a b, \quad e \perp a, \quad \|b\|^2 = \|p\|^2 + \|e\|^2.$$

Projection matrix:

$$P = \frac{aa^T}{\|a\|^2}, \quad P^2 = P, \quad P^T = P.$$

Orthonormal columns:

$$Q^T Q = I, \text{ and projection onto } \mathcal{C}(Q) \text{ is } P = QQ^T.$$

LEAST SQUARES

Normal equations:

$$A^T A \hat{x} = A^T b; \text{ residual } e = b - A\hat{x} \text{ satisfies } A^T e = 0.$$

$$\hat{x} = (A^T A)^{-1} A^T b, \quad p = A\hat{x} = Pb.$$

Gram-Schmidt:

$$q_1 = a_1 / \|a_1\|; \quad v_k = a_k - \sum_{j < k} (q_j^T a_k) q_j, \quad q_k = v_k / \|v_k\|.$$

QR factorization:

$$A = QR, \text{ where } Q^T Q = I \text{ and } R \text{ is upper triangular.}$$

PROBLEM SET

Q1.

Let

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \end{bmatrix}.$$

- (i) Find a basis and the dimension of each of the four fundamental subspaces of A .
- (ii) Verify directly that the row space is perpendicular to the nullspace, and that the column space is perpendicular to the left nullspace.

Q2.

Let $S \subseteq \mathbb{R}^4$ be spanned by

$$s_1 = (1, 1, 0, 0), \quad s_2 = (0, 1, 1, 0).$$

- (i) Find a basis for $S^\perp$.
- (ii) Find $(S^\perp)^\perp$ and compare it with S .

Q3.

Let $a = (1, 2, 2)$ and $b = (2, 1, 1)$.

- (i) Find the projection p of b onto the line through a , and find the error $e = b - p$.
- (ii) Verify that $e \perp a$ and that $\|b\|^2 = \|p\|^2 + \|e\|^2$.
- (iii) Find the projection matrix P onto the line through a , and verify that $P^2 = P$.

Q4.

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

- (i) Use the normal equations $A^T A \hat{x} = A^T b$ to find the projection $p = A \hat{x}$ of b onto $\mathcal{C}(A)$.
- (ii) Find the error $e = b - p$ and verify that it is perpendicular to both columns of A .
- (iii) Find the projection matrix $P = A(A^T A)^{-1} A^T$.

Q5.

Let A have independent columns and define

$$P = A(A^T A)^{-1} A^T.$$

- (i) Prove that $P^T = P$ and $P^2 = P$.
- (ii) Show that Pb is the projection of b onto $\mathcal{C}(A)$ by proving that $Pb \in \mathcal{C}(A)$ and $b - Pb \in \mathcal{N}(A^T)$.
- (iii) Show that $I - P$ is the projection matrix onto $\mathcal{N}(A^T)$.

Q6.

Fit the data $(t, b) = (0, 1), (1, 2), (2, 2)$ by the least-squares line $b = c + dt$.

- (i) Write the system as $A \begin{bmatrix} c \\ d \end{bmatrix} \approx b$ and form the normal equations.
- (ii) Find $\hat{c}$ and $\hat{d}$, and hence find the fitted values p .
- (iii) Find the residual $e = b - p$ and verify that it is perpendicular to both columns of A .

Q7.

Apply Gram-Schmidt to the columns

$$a_1 = (1, 1, 0), \quad a_2 = (1, 0, 1), \quad a_3 = (0, 1, 1).$$

- (i) Find an orthonormal basis q_1, q_2, q_3 .
- (ii) Write the matrix $A = [a_1 \ a_2 \ a_3]$ as $A = QR$, where $Q = [q_1 \ q_2 \ q_3]$.

Q8.

Let

$$q_1 = \frac{1}{\sqrt{2}}(1, 1, 0), \quad q_2 = \frac{1}{\sqrt{6}}(1, -1, 2),$$

and let $Q = [q_1 \ q_2]$. For $b = (2, 1, 3)$,

- (i) Find the coordinates $Q^T b$ and the projection $p = QQ^T b$ of b onto $\mathcal{C}(Q)$.
- (ii) Find the error $e = b - p$ and verify that $Q^T e = 0$.